Executive Summary

Water scarcity is a growing concern in semi-arid regions of the United States, particularly in the Western states, where agriculture depends heavily on groundwater and irrigation systems. Small and mid-sized farms often lack access to affordable and practical digital tools that support water-efficiency planning and resource optimization. As climate variability intensifies and groundwater reserves decline, improving agricultural water management has become both an environmental and economic priority.

The BioLoop Water Efficiency Model(BWEM) is a digital decision-support tool designed to assist small and mid-sized agricultural producers in estimating water demand, identifying inefficiencies, and implementing water-saving strategies. The system integrates farm-specific inputs, including land size, livestock population, crop type, reference evapotranspiration (ETo), and irrigation method, to generate structured water-use assessments and efficiency recommendations.

The goal of this initiative is to improve on-farm water planning through accessible, data-driven modeling. By enhancing water-use efficiency at the farm level, the tool aims to contribute to groundwater conservation, climate resilience, and long-term agricultural sustainability in water-stressed regions of the United States.

The initial phase focuses on model development, structured output generation, and pilot validation. Future expansion may integrate broader circular resource management features within the BioLoop framework.

Background and Problem Statement

Water availability has become one of the most critical environmental and economic challenges in semi-arid regions of the United States. States such as Colorado, Arizona, New Mexico, Nevada, and parts of California experience recurring drought conditions, declining snowpack, and increasing groundwater depletion. Agriculture remains one of the largest consumers of freshwater resources in these regions, accounting for a significant percentage of total water withdrawals.

While large-scale agricultural operations often have access to advanced irrigation management systems and technical consultants, small and mid-sized farms frequently rely on traditional estimation methods or generalized recommendations. Many producers do not have access to practical digital tools that translate climate data, livestock demand, and crop requirements into clear, farm-specific water planning insights.

In livestock production systems, water demand is influenced by animal type, body weight, environmental temperature, and production stage. Similarly, crop irrigation needs vary depending on soil type, evapotranspiration rates, and irrigation efficiency. Without structured modeling support, water use may be either underestimated, leading to productivity loss, or overestimated, contributing to waste and groundwater stress.

In semi-arid agricultural regions, inefficient irrigation practices and limited data integration can accelerate aquifer depletion and reduce long-term water security. As climate variability increases, the need for scalable, accessible water-efficiency planning tools becomes more urgent.

Currently, there is a gap between available scientific knowledge on agricultural water requirements and the practical decision-making tools accessible to small and mid-sized producers. A structured, user-oriented digital model designed specifically for water-stressed agricultural systems can help bridge this gap.

These conditions are well documented in U.S. irrigation and drought monitoring sources **(see References).**

(BioLoop Water Efficiency Model – BWEM)

3. Proposed Solution: BioLoop Water Efficiency Model (BWEM)

The BioLoop Water Efficiency Model (BWEM) is a digital decision-support platform designed to improve water-use planning for small and mid-sized agricultural producers operating in semi-arid regions of the United States. The system integrates farm-specific variables into a structured modeling framework to generate water demand estimates and efficiency recommendations.

3.1 System Inputs

The model collects key operational and environmental data, including:

- Farm size (acres or hectares)
- Crop type and growth stage
- Livestock species and population size
- Average animal body weight (if applicable)
- Reference evapotranspiration (ETo) value (user-entered; sourced from regional weather station datasets such as CoAgMET).
- Irrigation method (flood, sprinkler, drip, etc.)
- Soil type (optional in Phase 1)

These inputs allow the system to calculate water demand at both the crop and livestock levels.

3.2 Core Modeling Components

The BWEM integrates:

• Crop Water Requirement Estimation

Based on evapotranspiration (ET) modeling and crop coefficient (Kc) values derived from established agricultural standards.

• Livestock Water Demand Calculation

Species-based daily intake assumptions (planning-level), with optional temperature-based adjustment where applicable.

• Irrigation Efficiency Assessment

Comparison between estimated optimal water need and reported irrigation application.

• Water Stress Sensitivity Adjustment (Phase 1 simplified)

Adjustment factor for high-temperature or drought conditions.

3.3 System Outputs

The model generates:

- Estimated total farm water demand
- Breakdown by crop vs. livestock consumption
- Irrigation efficiency percentage
- Identified efficiency gaps
- Recommended water-saving strategies
- Scenario comparison (optional in Phase 1)

The output is structured in clear, visual summaries to support decision-making at the farm level.

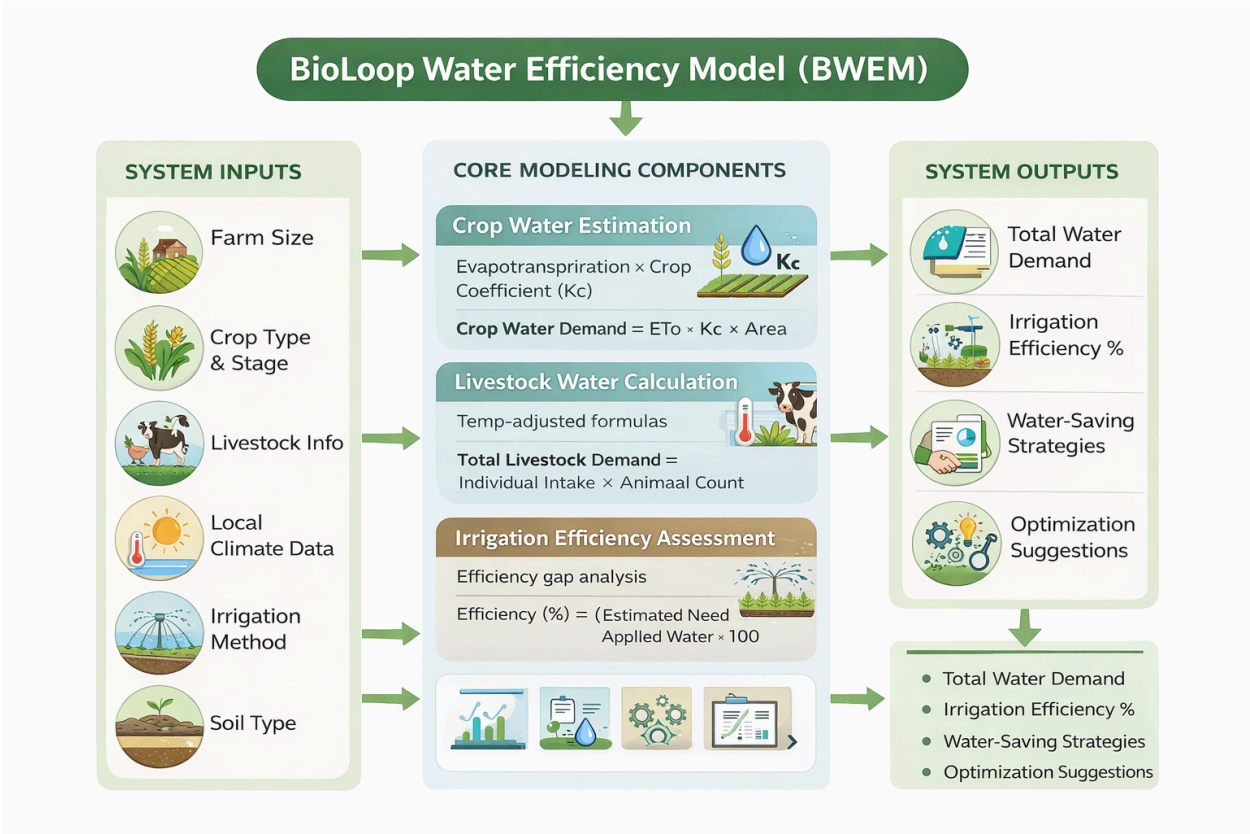
3.4 Accessibility & Scalability

The platform is designed to be:

- Web-based
- Low-cost
- User-friendly
- Scalable across multiple U.S. semi-arid states

The modular design allows future expansion into broader circular resource management systems within the BioLoop framework.

Diagram:



- Note: BWEM converts ETc depth to volumetric demand using irrigated area and standard unit conversions (mm → liters → gallons), and then adjusts applied irrigation by irrigation system efficiency (see E3 for calculation details).

4. Methodology and Data Sources

4.1 Modeling Approach

The BioLoop Water Efficiency Model (BWEM) uses a structured, data-informed modeling framework that integrates established agricultural water estimation principles with publicly available climate and livestock consumption data. The system is designed as a decision-support tool rather than a predictive simulation platform, focusing on practical and scalable water-efficiency assessment.

The modeling approach combines three core components:

1. Crop water requirement estimation using evapotranspiration-based calculations

2. Livestock water intake estimation based on species-specific standards

3. Irrigation efficiency assessment comparing estimated optimal demand with reported application

This modular structure allows the model to remain scientifically grounded while adaptable to different farm configurations.

4.2 Data Sources

The BWEM framework relies on publicly accessible and scientifically validated data sources, including:

• Reference evapotranspiration (ETo) inputs are sourced from regional weather station networks (e.g., CoAgMET for Colorado) and other publicly available station datasets.

• Crop coefficient (Kc) values are sourced from FAO-56 and university/extension irrigation scheduling references.

• Livestock water intake assumptions are supported by extension-based planning references (e.g., CSU Extension cattle water requirement guidance)

• Irrigation efficiency benchmarks from USDA Natural Resources Conservation Service (NRCS) and state-level water conservation agencies

By using standardized national data sources, the model ensures consistency, transparency, and scalability across multiple semi-arid regions of the U.S.

4.3 Assumptions and Simplifications (Phase 1)

To ensure usability and accessibility for small and mid-sized farms, Phase 1 of the BWEM incorporates several controlled assumptions:

• Average seasonal evapotranspiration values are used where real-time weather integration is not available

• Livestock water intake is estimated using temperature-adjusted average formulas rather than continuous physiological monitoring

• Soil variability is incorporated at a general classification level rather than site-specific soil hydrology modeling

These simplifications allow the model to provide practical planning-level estimates while maintaining computational efficiency.

4.4 Scalability and Future Integration

The BWEM architecture is modular and designed for future enhancement. Potential future integrations include:

• Real-time weather API integration
• Remote sensing data incorporation
• Soil moisture sensor compatibility
• Expanded circular resource modeling within the BioLoop framework

This staged development strategy ensures the initial implementation remains feasible while preserving long-term scalability.

4.5 Pilot Testing and Validation Strategy

BWEM has already been tested through pilot scenario simulations using region-specific Colorado reference ETo inputs and crop coefficients across representative semi-arid farm profiles. These pilots include irrigated alfalfa and irrigated corn scenarios with differing irrigation efficiencies (e.g., flood vs drip; sprinkler) and optional dairy water-demand components, demonstrating operational feasibility and sensitivity to key parameters.

Next validation steps will expand testing to additional datasets and production contexts, compare outputs against published benchmarks, incorporate stakeholder feedback, and refine assumptions and interface options as the tool is prepared for broader deployment.

4.5.1 Pilot Implementation

Phase 1 Expanded validation will involve:

• Simulated farm case studies using publicly available agricultural datasets
• Application of the BWEM model to hypothetical small and mid-sized farm scenarios in semi-arid regions (e.g., Colorado, Arizona, New Mexico)
• Comparative assessment against published regional water-use benchmarks

These simulated case studies allow the system's outputs to be evaluated against established agricultural water standards before field deployment.

4.5.2 Benchmark Comparison

Model outputs will be compared with:

• USDA irrigation water-use estimates
• State-level agricultural water planning guidelines
• University extension livestock water requirement standards

Deviation analysis will identify whether BWEM outputs fall within scientifically accepted water demand ranges.

4.5.3 Future Field Validation (Phase 2)

Following initial simulation validation, the next phase may include:

• Collaboration with small-scale agricultural producers or university extension programs
• Collection of anonymized farm water-use data
• Refinement of efficiency thresholds based on observed application rates

This staged validation approach ensures that the BWEM evolves through structured testing rather than speculative modeling.

4.5.4 Risk and Limitation Transparency

The validation process acknowledges that:

• Regional variability in soil and microclimate conditions may affect precision
• The tool is intended to support planning decisions, not replace professional hydrological assessments
• Continuous improvement will depend on data refinement and user feedback

Including validation strategy demonstrates technical seriousness and practical feasibility.

4.5.5 Limitations and Future Development

The current BioLoop Water Efficiency Model (BWEM) prototype is designed as a planning-level assessment tool intended to provide structured estimates of farm-level water demand and irrigation efficiency. While the model integrates established evapotranspiration methodology, livestock water demand assumptions, and irrigation efficiency benchmarks, several limitations are acknowledged.

First, the model relies on standardized crop coefficients and generalized climate inputs rather than site-specific soil moisture monitoring or real-time meteorological integration. As such, results are intended to support comparative planning and decision-making rather than precise agronomic prescriptions.

Second, irrigation efficiency values are based on typical system performance assumptions (e.g., flood, sprinkler, drip) and may vary depending on local infrastructure, maintenance practices, and field conditions.

Third, livestock water intake estimates are calculated using representative daily consumption averages and do not dynamically adjust for production stage, humidity, or breed-specific variation.

Key Limitations (Current Prototype Version)

- **Prototype scope:** Crop options are currently limited to the crop types included in the prototype interface, and livestock categories are limited to the types included in the prototype interface.
- **Irrigation methods:** Irrigation methods are currently limited to **flood, sprinkler, and drip**. In the current version, **"drip" is treated as a general category and does not yet differentiate surface drip versus subsurface drip**.
- **Planning-level model:** BWEM is intended for planning-level estimates and scenario comparisons; it does not currently model effective precipitation, soil moisture storage dynamics, runoff/return flows, or field-scale irrigation scheduling constraints.
- **Water quality/source:** Water quality considerations and water source constraints (e.g., testing/treatment requirements) are outside the current model scope.

Future development phases may include:

• Integration of real-time climate data through weather APIs
• Incorporation of soil classification parameters and moisture variability
• Enhanced livestock modeling adjustments for temperature stress
• Scenario comparison tools for irrigation system transition analysis
• Expansion into broader circular agricultural resource optimization modules within the BioLoop framework

These limitations do not diminish the functional value of the current prototype but instead reflect a phased and structured development strategy focused on technical feasibility, scalability, and incremental refinement.

5. Prototype Development and Implementation Status

A functional planning-level prototype of the BioLoop Water Efficiency Model has been developed and deployed through a publicly accessible web platform. The prototype integrates crop evapotranspiration modeling, livestock water demand estimation, and irrigation efficiency analysis into a structured decision-support tool.

The current implementation allows users to:

- Input farm area, crop type, and local climate reference evapotranspiration (ETo)
- Select irrigation method efficiency (e.g., flood, sprinkler, drip)
- Enter livestock type and population size
- Generate daily water demand estimates for crops and livestock
- Calculate applied water requirements and estimated irrigation efficiency
- Identify potential water losses and receive basic water management recommendations

The model calculates crop water requirements using standard evapotranspiration methodology ($ETc = ETo \times$ crop coefficient), integrates livestock daily water intake assumptions based on species type, and adjusts total applied water based on irrigation system efficiency factors.

The prototype is designed as a planning-level assessment tool intended to help agricultural producers better understand water demand patterns and irrigation efficiency in semi-arid environments. While not a substitute for site-specific agronomic consultation, the tool demonstrates the feasibility and operational structure of the BioLoop framework.

This implementation represents the initial phase of the broader BioLoop system, which aims to expand into integrated resource optimization across water, nutrient cycling, and circular agricultural systems.

6. Expected National Impact

6.1 Substantial Merit in Environmental and Agricultural Sustainability

Water security is directly linked to environmental stability, agricultural productivity, and regional economic resilience in the United States. Semi-arid regions, particularly in the Western states, face increasing pressure from drought cycles, declining snowpack, groundwater depletion, and rising agricultural demand.

Agriculture remains a major water-consuming sector in these regions. Improving water-use efficiency at the farm level contributes directly to:

- Groundwater conservation
- Reduced pressure on overallocated river systems
- Increased climate resilience in rural communities
- Long-term agricultural sustainability

By providing a structured and accessible digital decision-support tool, the BioLoop Water Efficiency Model (BWEM) addresses a documented gap between scientific water management knowledge and practical implementation at the small and mid-sized farm level.

This establishes substantial merit in environmental protection, sustainable agriculture, and resource management.

6.2 National Importance Beyond a Single Employer

The BWEM is not designed for a single company, employer, or private entity. The model is structured to be:

- Scalable across multiple semi-arid states
- Adaptable to diverse crop and livestock systems
- Accessible to independent agricultural producers

Water scarcity is not confined to one state or region. Federal and state-level policies increasingly emphasize water conservation, drought mitigation, and climate adaptation strategies.

A scalable water-efficiency planning tool aligns with national priorities related to:

- Climate adaptation planning
- Agricultural sustainability initiatives

• Groundwater management reform efforts
• Federal conservation funding programs

Because the model is broadly applicable across agricultural systems in water-stressed regions, its potential impact extends beyond localized operations and supports national-level resource resilience.

6.3 Economic and Community-Level Impact

Small and mid-sized farms represent a significant portion of agricultural producers in the United States. Many of these producers operate with limited access to precision resource planning technologies.

By improving water-use estimation and efficiency planning, the BWEM can contribute to:

• Reduced operational costs
• Improved crop and livestock productivity
• Can help increase long-term farm viability
• Can contribute to rural economic stability

In drought-prone regions, water mismanagement can lead to crop failure, livestock stress, and financial loss. Decision-support tools that reduce uncertainty can mitigate these risks.

6.4 Alignment with Federal Sustainability Priorities

Federal agencies, including the USDA, Bureau of Reclamation, and climate-focused programs, emphasize water conservation and sustainable agricultural practices.

The BWEM framework supports these national objectives by:

• Promoting data-driven water allocation decisions
• Encouraging efficient irrigation practices
• Supporting conservation-based agricultural planning

This alignment reinforces the national relevance of the proposed endeavor.

6.5 National Scope and Practical Advancement

The proposed endeavor advances a field of substantial importance to the United States by addressing water-efficiency challenges in agriculture through a scalable and data-informed modeling approach. The initiative is not limited to a specific employer or commercial contract but is structured for broad application across water-stressed agricultural regions. The model's design prioritizes accessibility, adaptability, and phased validation, demonstrating that the proposed work is both practically achievable and positioned to contribute meaningfully to national sustainability objectives.

The phased development strategy further reflects a realistic implementation plan, beginning with a focused water-efficiency framework and expanding through validated enhancements. This structured progression supports long-term advancement in sustainable agricultural resource management while maintaining technical feasibility.

7. Founder Qualifications and Positioning

The BioLoop Water Efficiency Model (BWEM) is led and developed by Khadidja Moussouni, whose academic and applied background integrates animal production, agricultural systems, and environmental science.

Ms. Moussouni holds a Master's degree in Animal Production and Nutrition, with academic training focused on livestock systems, resource management, and production efficiency. Her graduate research experience included the development of a structured application model designed to optimize feeding strategies in dairy cattle systems. Through this work, she gained practical experience in modeling biological requirements, integrating data inputs, and translating agricultural science into applied decision-support tools.

In addition to her background in animal production, Ms. Moussouni is pursuing advanced graduate study in Environmental Sciences in the United States. Her coursework and academic focus include biogeochemical cycles, water systems, and sustainability frameworks, strengthening her interdisciplinary capacity to address agricultural water management challenges.

This combined expertise uniquely positions her to bridge livestock production knowledge with environmental resource modeling. The BWEM framework reflects this integration by incorporating species-specific livestock water demand calculations alongside crop evapotranspiration modeling within a unified system.

Her academic progression from animal production systems to environmental resource analysis demonstrates a focused and evolving commitment to sustainable agricultural practices. The development of BWEM represents a continuation of her prior applied modeling work, expanding from feed optimization to water-efficiency optimization within semi-arid agricultural systems.

Through this interdisciplinary background and prior application development experience, Ms. Moussouni is well positioned to advance the proposed endeavor in a structured and technically grounded manner.

9. Conclusion

Water scarcity presents a growing environmental and economic challenge in semi-arid regions of the United States, where agricultural production depends heavily on efficient water allocation. Small and mid-sized farms often operate without accessible, structured decision-support tools that translate scientific water-use principles into practical planning strategies.

The BioLoop Water Efficiency Model (BWEM) addresses this gap through a data-informed, scalable digital framework designed to estimate farm-level water demand, evaluate irrigation efficiency, and generate structured resource optimization recommendations. By integrating crop evapotranspiration modeling, livestock water demand estimation, and irrigation efficiency assessment within a unified system, the model provides actionable insights to support sustainable agricultural practices.

The phased implementation strategy, pilot validation plan, and reliance on established national data sources demonstrate the technical feasibility and structured progression of the proposed endeavor. The model's scalable design further enables expansion across multiple semi-arid agricultural regions.

Through its focus on practical water-efficiency optimization, the BWEM initiative contributes to broader objectives of groundwater conservation, climate resilience, and long-term agricultural sustainability in the United States.

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